LI-7700 methane gas analyzer maintenance

Important: Power off the gas analyzer (disable the power supply or disconnect the power cable) before conducting any maintenance that involves disassembly or disconnecting cables. Disassembling the instrument or detaching or attaching head cables while the instrument is powered on may damage the instrument or result in data loss.

Before performing maintenance to the sensor, please check thoroughly the details of that procedure in this document, if any questions arise do not hesitate to contact David Trejo (trejo_cancino.david_ulises@uqam.ca).

After conducting any maintenance protocol, record it in the EcoFlux Inventory notes to have a follow-up of each sensor and station. If something is wrong with any equipment, please contact David Trejo Cancino (trejo_cancino.david_ulises@uqam.ca) to register it in the database and proceed with the required diagnostics and repairs.

Summary

Procedure	Frequency	Items required
Changing the desiccant	Once a year in humid environments and when *Optic Rh > 30%*.	- Desiccant bottle. - Pliers. - PH#1 screwdriver.
Cleaning the mirrors / filling the water reservoir	Every site visit. When signal strength is low.	- Demineralized water. - Microfiber soft cloth. - Windshield washer fluid.
Changing the thermocouple	Replace if the wire is broken or there are wrong readings.	- PH#1 screwdriver. - Thermocouple.
Changing the LI-7550 fuses (2)	If fuses are blown. Check resistance, it should be < 1 Ohm.	- Flat-blade screwdriver. - Power supply fuse. - 2A Nano2 SMF Fuse.
Changing the LI-7700 fuse	If the LI-7700 fails to power up. If the fuse is blown.	- 8A FAST 250V TR5 UL Fuse.
Update software and embedded firmware	When there is a new release previously tested in the lab.	- Ethernet cable. - Software LI7700_updater.exe.
User calibration: zero and span	Check readings. At least every 6 months. Calibrate when the temperature setting has been changed	- Zero gas. - CH_4 gas. - Connections between the gas tanks and the sensor. - Flowmeter 0.5-1 LPM.
Changing the operating temperature range	Before/after winter.	- Software LI7xx0. - Items for calibration.

Changing the internal desiccant bottle

There is one internal desiccant bottle in the LI-7700. It is filled with Blue-Indicating Drierite®. The desiccant cap is above the circular plug located on the bottom of the top dome of the analyzer (see below). The desiccant bottle should be changed annually when the LI-7700 is used in humid environments. Desiccant bottles and recharge kits are available from LI-COR (part number 7700-950).

In the Calibration section of the Dashboard, observe the field that reads *Optics RH*. Normally this should be near zero. If the indicator turns from green to red (indicating the optics RH: >30%), then it is time to change the desiccant. To change the desiccant:

1) **Remove the set screw that secures the desiccant plug**
   If it is not in place, insert the knurled screw from the spares kit (part number 125-09760) into the desiccant plug. Grip the screw with your fingers or a pair of pliers and pull straight out to remove the desiccant plug.





2) **Remove the seal from the top of the replacement desiccant bottle**
   Insert the bottle cap first and replace the foam disk and cover. Press firmly to secure the desiccant cover and replace the set screw.



Note: A droplet of saliva or dish soap applied to the O-rings can ease the insertion of the desiccant cover.

Over time, the Optics RH: should approach zero. This may take several minutes to several hours, depending on the humidity level. There should be no need to perform any other maintenance procedures (e.g., zero or span) after this.

Cleaning the mirrors

Important: Never use abrasive cleaners on the mirror. Refrain from using polishing compounds or toothpaste. These can permanently damage the mirror finish, which will require costly repairs.

The upper and lower mirrors are extremely scratch-resistant; however, the mirror surfaces should be treated the same way you would treat the surface of an expensive lens. In general, refrain from applying a lot of pressure when cleaning a dry mirror. If the mirror is dusty, just wipe it with a soft, clean, moistened cloth. If a sticky substance, such as pollen, has built up on the mirror, a mild soap or a commercial glass cleaner.

Note: Products such as Rain-X® can be applied to the lower mirror to help the glass shed water.

There may be the appearance of “spots” on the mirrors. These are normal and are the result of the manufacturing process. The spots are in the visible spectrum and will not affect reflectance near 1.6 microns, which is where the LI-7700 operates. The washer reservoir uses standard over-the-counter windshield washer fluid. If the fouling of the mirror is minor, tap water will work fine. Cleaners that contain methylalcohol (methanol) and isopropyl alcohol are suitable. Refrain from using cleaners that contain powerful solvents such as acetone. As a general rule, if the cleaner is safe for automotive finishes, it is probably safe to use in the LI-7700 washer reservoir.

The washer should be filled with windshield washer fluid with a temperature rating suitable for the environment in which it will be deployed. For highly sensitive environments, select an environmentally friendly solution.

If your instrument has persistent noise, there may be invisible water deposits on the lower mirror. You may need to clean it more thoroughly. If needed, position the instrument vertically and flood the lower mirror with white vinegar. Allow it to sit for an hour and then rinse thoroughly with distilled water. Be sure that the mirror is free of solids that can scratch the surface, and then gently polish it with a soft cloth.

Changing the thermocouple

Important: Power off the gas analyzer (disable the power supply or disconnect the power cable) before conducting any maintenance that involves disassembly or disconnecting cables.

The fine wire thermocouple should be replaced if the wire is broken or if it reads incorrectly. One spare thermocouple is included in the spares kit, and others can be purchased from LI-COR, (part number 9977-038). To replace the thermocouple:

1) **Fully loosen the two screws (#1 Philips) that secure the thermocouple assembly to the bottom of the upper housing.**
The screws will be retained in the thermocouple base.



2) **Grasp the thermocouple housing and pull it straight out.**
It should slide out easily.

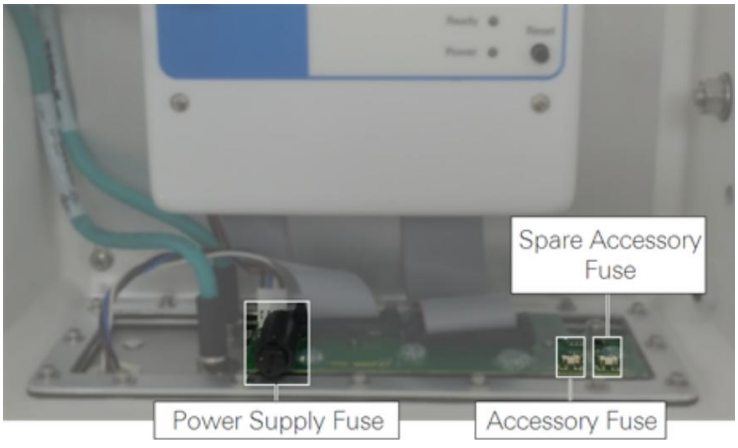


3) **Insert the new thermocouple.**
The thermocouple has a protective plastic sleeve - leave it in place. Be sure the bend is directed toward the optical path. Handle the thermocouple carefully to protect the fine wires.
4) **Replace the two screws.**
Connect with the LI-7700 and verify that the temperature readings are correct.

Changing the LI-7550 fuses

Important: Power off the gas analyzer (disable the power supply or disconnect the power cable) before conducting any maintenance that involves disassembly or disconnecting cables.

The LI-7550 has two fuses – one for the power supply and one for the accessory. To check a fuse, inspect it for evidence that it is burned, and use an ohm meter to check the resistance. Resistance of <1 Ω (ohm) indicates that the fuse is OK.



Power Supply Fuse

The power supply is protected by a 5 A 125/250V, 5 × 20 mm fast-blow fuse (part number 439-04214). If the battery or other power source fails to power the instrument, check to see if the fuse has blown.

The fuse is in the lower left-hand corner, as shown in the figure.

Replacement fuses (part number 439-04214) are in the spares kit. Use a flat-blade screwdriver or your thumb to push down on the fuse holder cap and turn counterclockwise to release the cap.

Accessory Fuse

The accessory uses a 2A Nano^2 SMF Fuse. There is one spare fuse included in the LI-7550. If the heated intake tube will not power on or continuously triggers an error, check the fuse and replace it if necessary.

Changing the LI-7700 fuse

Important: Power off the gas analyzer (disable the power supply or disconnect the power cable) before conducting any maintenance that involves disassembly or disconnecting cables.

If the LI-7700 fails to power up when connected to a suitable power supply, check to see if the fuse has blown. The LI-7700 fuse is enclosed within the knurled fuse cover on the connection panel. One extra fuse (8A FAST 250V TR5 UL; part number 438-15034; early models used a 5-amp fuse. Contact LI-COR if you have questions) is included in the spares kit. To replace the fuse, disconnect the power supply to power off the LI-7700, remove the fuse cover, and pull out the old fuse. Insert the replacement fuse and reattach the fuse cover.



Updating the software

The latest software is available from licor.com/7700software. Updating the Windows interface software is a simple matter of downloading the latest version and installing it on your computer. Updating the embedded firmware involves downloading the embedded firmware updater, installing it on your computer, and loading the new firmware onto the instrument.

1) **Connect the LI-7700 to your computer or local network using the Ethernet cable.**
2) **Open the file called li7700_updater.exe.**
3) **Select the LI-7700 from the list and click Update Firmware.**

User calibration: zero and span

The implementation of the zero and span into calculations is given in the equation summary. The overall accuracy of the LI-7700 depends upon its zero and span calibration. Always use quality zero and span gases in a balance of air with CH4 accuracy greater than 1%. Use gases that are free of Volatile Organic Compounds (VOCs) other than methane. Check the readings every 6 months.

Performing a zero calibration

To zero the LI-7700:

1) **Connect the LI-7700 to a power supply and a computer.**
   Launch the LI-7700 software and connect with the instrument.
2) **Remove the radiation shield.**
   The radiation shield will fit on very tightly. Do not yank on it or apply excessive force.
3) **Install the calibration shroud.**
   Ensure that it seals around the top and bottom openings.
4) **Connect the zero gas (0 ppm CH_4 in air) to the shroud fittings.**
5) **Flow the gas through the shroud.**
   Depending on the flow rate, allow from 10-30 minutes for equilibration.
6) **Click the *Zero CH_4* button in the calibration frame.**
   When the instrument begins the zero operation, the panel will update to reflect the activity. The Zero CH_4 will change to Abort. Clicking Abort terminates the zero procedure. It usually takes about 10 seconds to zero. If it takes significantly longer, check for leaks and verify the calibration tank pressure.
7) **After the software completes the zero operation choose either to apply the new calibration value or revert to the previous calibration value.**
   If the application is closed or if the LI-7700 is powered off before a confirmation command is received the new calibration will not be applied.

Performing a span calibration

Performing a span is essentially the same as a zero, except that you must flow the span gas through the optical path and enter the span gas concentration in parts per million (ppm). Select a span gas with a methane concentration at or slightly above the methane concentrations you expect to measure. For the measurement of methane in ambient air, 2 ppm methane in a balance of air is ideal.

1) **Connect the span gas (around 2 ppm CH_4 in air) to the shroud fittings.**
2) **Type the concentration (ppm) of the span gas into the field on the calibration pane.**
3) **Flow the gas through the shroud.**
   Allow from 10 to 30 minutes for equilibration, depending on flow rate.
4) **Click the *Span CH_4* button in the calibration frame.**

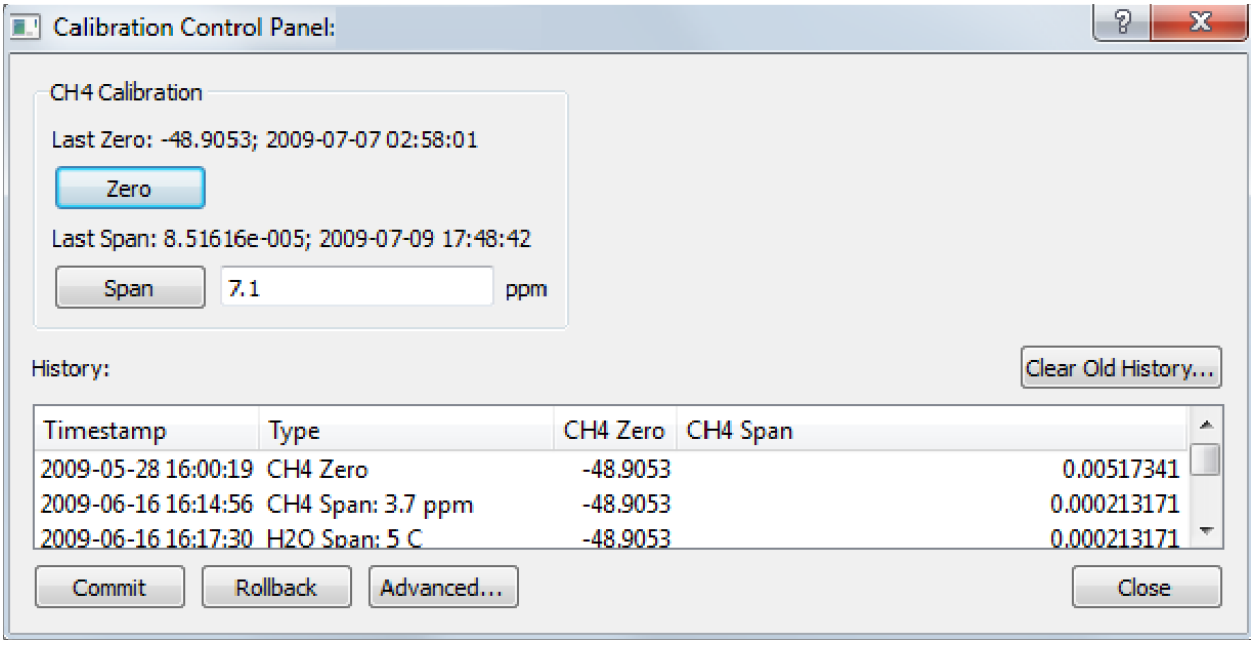
When the instrument begins the zero operation, the panel will update to reflect the activity. The *Span CH_4* will change to *Abort*. Clicking *Abort* terminates the zero procedure. It usually takes about 10 seconds to zero. If it takes significantly longer, check for leaks and verify the calibration tank pressure.

5) **After the software completes the zero operation choose either to apply the new calibration value or revert to the previous calibration value.**

Restoring a previous calibration

Should the calibration need to be restored to the factory's original calibration or any other prior calibration; this is accomplished through the *LI-7700 Finder Application*.

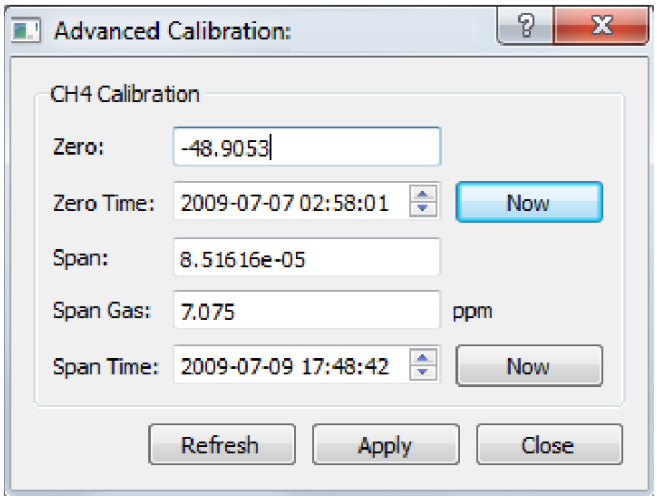
Launch the LI-7700 software, click the help question mark > About > Factory Setup > Proceed. Select your LI-7700 from the list and select Watch LI-7700. In the next window, click Cal…. This will display the calibration history for the instrument.



Make a note of the time stamp and write down the *CH_4 Zero* and *CH_4 Span* values associated with it with the previous calibration. Click the *Advanced…* button.

Enter the *CH_4 Zero* into the Zero field, then enter the *CH_4 Span* into the Span field.

Click both Now buttons to record the time that you entered these values. Close each of the open windows and verify the readings provided by the LI-7700.

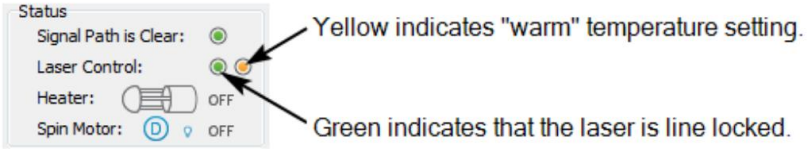


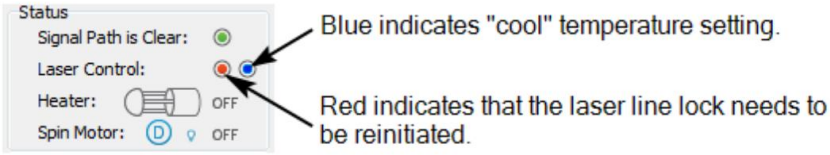
Changing the operating temperature range

Important: After changing the operating temperature range, be sure to re-zero and span the analyzer using high-quality (1% or better methane in air) calibration gases.

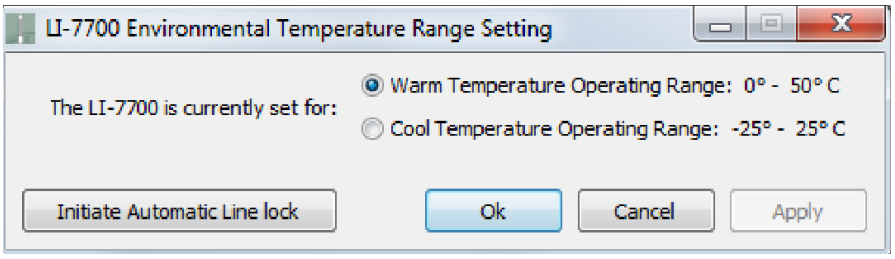
Two operating temperature ranges are available: -25 °C to 25 °C (cool) and 0 °C to 50 °C (warm). Select the temperature range setting that reflects the expected temperatures in the environment where you will deploy the LI-7700. If the wrong temperature setting is selected, the instrument may be unable to maintain the required temperature in the internal controls, and thus lose line lock (and it will not record methane data).

The current setting is indicated by either a blue (cool) or yellow (warm) indicator in the main window.





To change the setting, click on the yellow or blue indicator. This will open the window shown below, where you can change the setting.



After changing the temperature range setting, it may take up to 5 or 10 minutes for the block temperature to adjust before the instrument returns to normal operation. Look at the *Diagnostics Page 1* tab and observe the *Block Temperature* setting. *The Actual* and *Set* temperatures should be within about 1.5 °C after the temperature adjusts.

If the instrument is unable to achieve line lock once the new operating temperature range is established, click the Initiate Automatic Line Lock button in the temperature range setting window. This will instruct the instrument to re-try the automatic line lock routine.